

Schistosomes (Blood Flukes)

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- ❑ Schistosomes are dioecious trematodes in which the sexes are separates.
- ❑ The male is broader than females, and its lateral borders are rolled ventrally into a cylindrical shape , producing along groove or through called the gynaecophoric canal, in which the female is held.
- ❑ It appears as though the body of male is split longitudinally to produce thin canal-hence the name schistosome (greek schisto-split and soma body).

- ❑ Schistosomes were formerly called bilharzias after Theodor Bilharz who in 1851 , first observed the worm in the mesenteric veins of an Egyptian in Cairo.
- ❑ All schistosomes live in venous plexuses in the body of definitive host. The location varying with the species.

Schistosomes differ from the hermaphroditic trematoda in many respects:

- ❑ They lack a muscular pharynx.
- ❑ Their intestinal caeca reunite after bifurcation to form a single canal.
- ❑ They produce non-operculate eggs.
- ❑ They have no redia & metacercaria stages in larval development.
- ❑ The cercariae have forked tails and infect by penetrating the unbroken skin of definitive hosts.

- ❑ Schistosomiasis (bilharziasis) is a waterborne disease constituting an important public health problem affecting millions of persons in Africa, Asia and Latin America.
- ❑ It is estimated that over 200 million persons are infected with *S. haematobium*, *S. mansoni* , and *S. japonicum* each.
- ❑ Five species of schistosomes:
 1. *Schistosoma haematobium*
 2. *Schistosoma mansoni*
 3. *Schistosoma japonicum*
 4. *Schistosoma intercalatum*
 5. *Schistosoma mekongi*

Schistosoma haematobium:

History:

- ❑ This vesical blood fluke, formerly known as bilharzias haematobium has been endemic in the Nile valley in Egypt for millennia. its eggs have been found in the renal pelvis of an Egyptian mummies of predynastic period, 3100 B.C.
- ❑ The adult worm was described in 1851 by Bilharz in Cairo. Its life cycle including the larval stage in the snail was worked out by Leiper in 1915 in Egypt.

Geographical distribution:

- Although maximally entrenched in Nile valley *S. haematobium* is also endemic in most parts of Africa and in West Asia and middle East (Iraq) . An focus of endemicity in India was also identified.

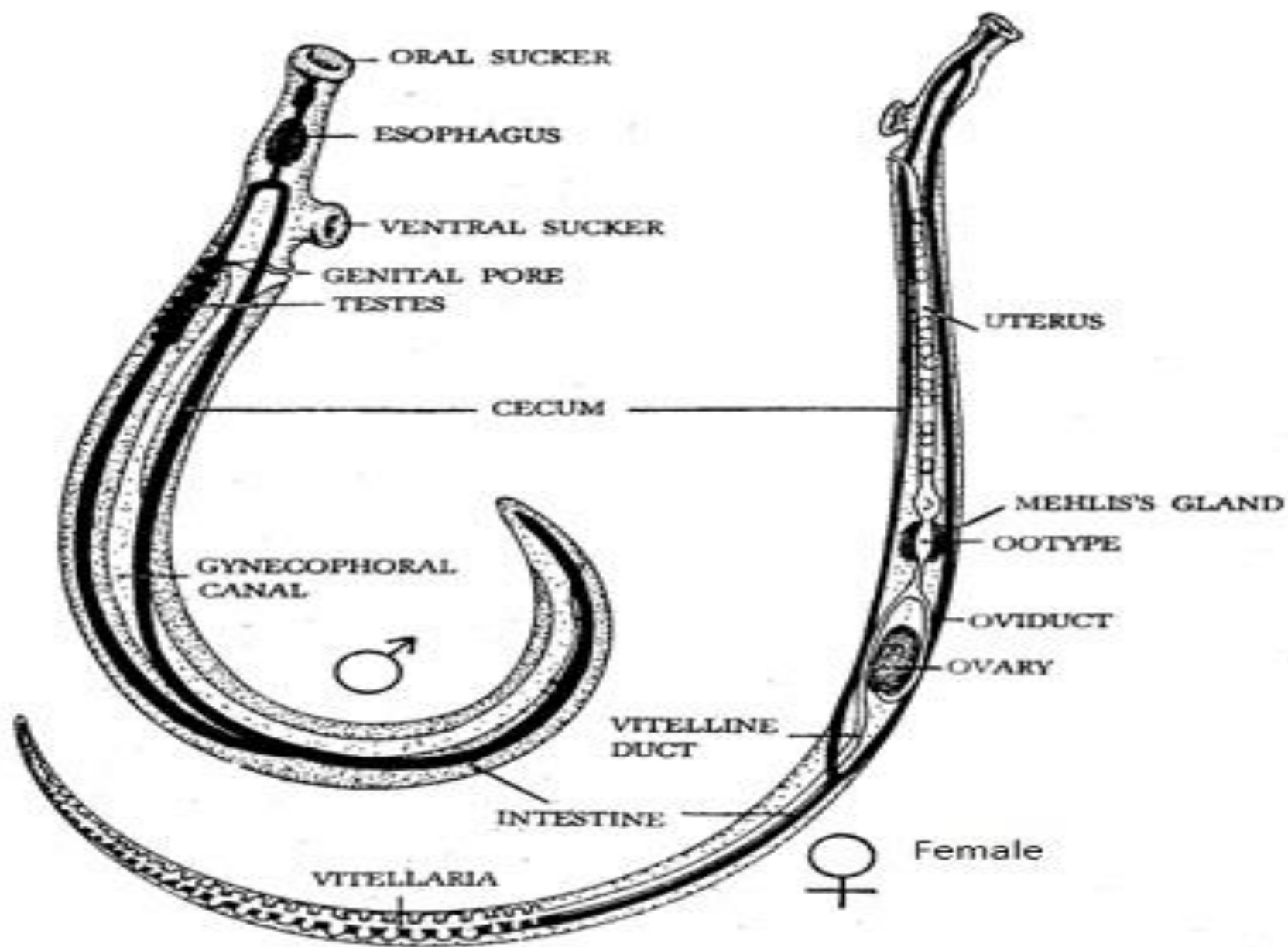
Morphology & Life cycle:

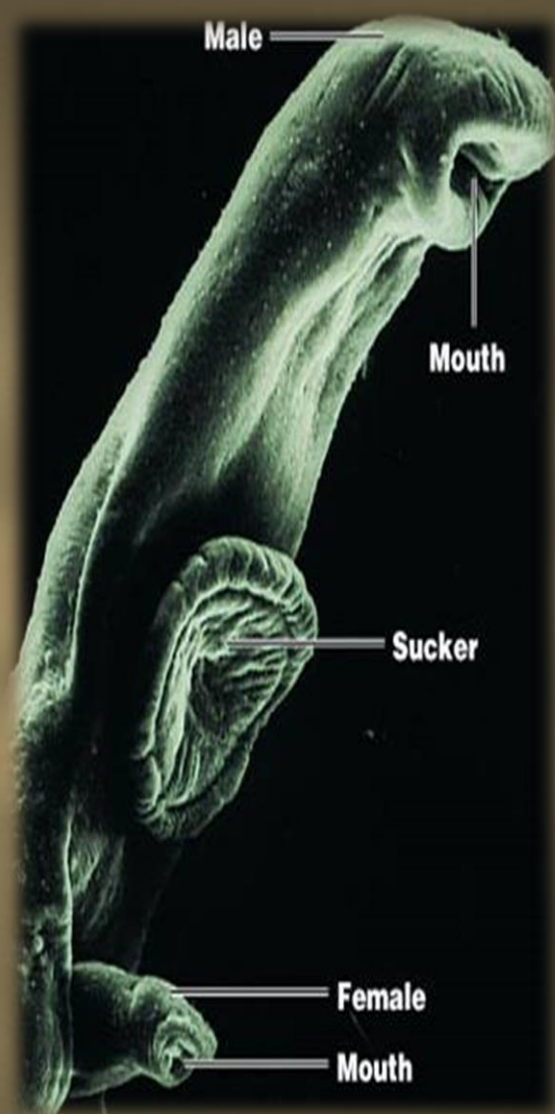
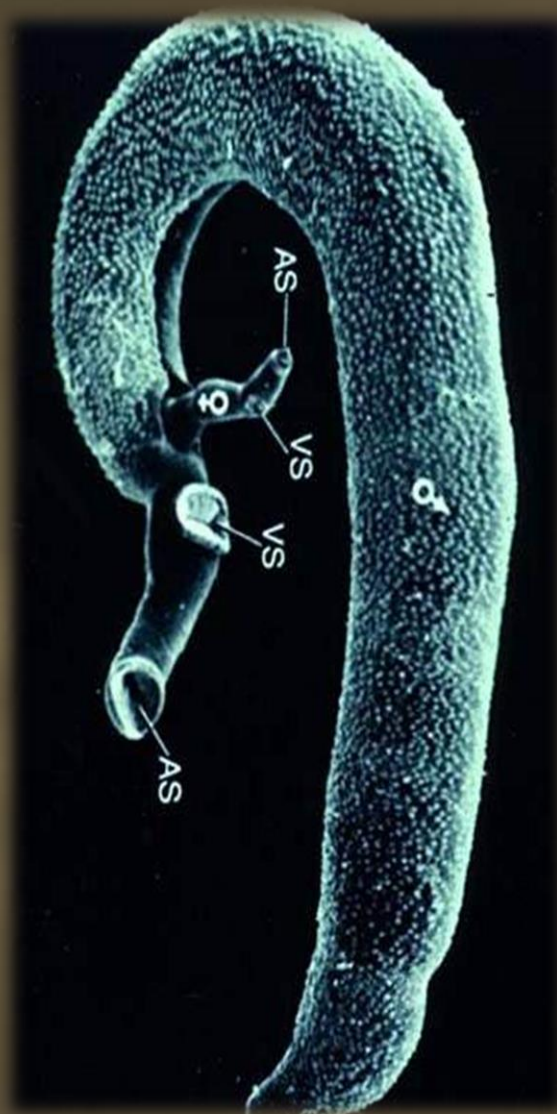
- The adult worms live in the vesical and pelvic plexus of veins. The male is 10-15 mm long by 1 mm thick and covered by a finely tuberculated cuticle.

- It has two suckers (muscular), the oral sucker being small and ventral sucker large and prominent. beginning immediately behind the ventral sucker and extending to caudal end is the gynaecophoric canal in which the female worm is held.
- Four to five small subglobose testes can be demonstrated close to one another behind the acetabulum. The two caeca are united in second half of the body. Genital pore locate behind the ventral sucker.

- ❑ The female is delicately cylindrical with length measurement of about 20 mm and diameter of 0.25 mm. the genital organs composed of ovary which is located in second half of the body from which extend ovi duct open in ootype.
- ❑ The uterus extend between genital pore (which located behind ventral sucker) and ootype , uterus contains about 20-30 eggs and vitellaria (yolk glands) which located in lateral field of united caeca in posterior half of the body.

Schistosomes: anatomy of the adults





Morphology (eggs)

S. mansoni



S. haematobium



S. japonicum



- ❑ The eggs are ovoid , about 150 μm by 50 μm with a brownish yellow transparent shell carrying a terminal spine at one pole (the terminal spine is characteristic of species).
- ❑ The eggs are laid usually in the small venules of the vesical and pelvic plexuses, through they are laid in the mesenteric portal system pulmonary arterioles and other ectopic sites.

□ The eggs are laid one behind the other with spine pointing posteriorly from venules the eggs make their way through the vesical wall by piercing action of spine assisted by mounting pressure within the venules and a lytic substance released by eggs.

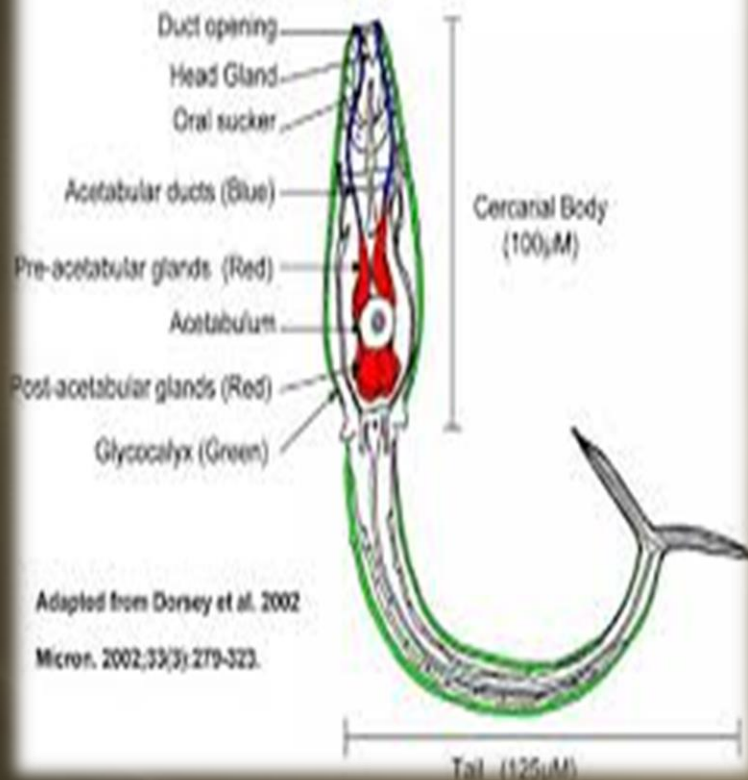
□ The eggs pass into the lumen of the urinary bladder together with some extravasated blood the eggs pass into urine, particularly towards the end of micturition.

- ❑ For some unknown reasons, the eggs are passed in urine mores during midday than at any other times of the day or night.
- ❑ The eggs laid in ectopic sites generally die and evoke local tissue reactions. They may be found , for instance in rectal biopsies but seldom passed live in feces.
- ❑ The eggs that are passed in water hatch releasing the ciliated miracidia.

- They swim about in water and on encountering a suitable intermediate host, penetrate into its tissues and reach its liver. The intermediate hosts are snails of *bulinus* species (*Bulinus truncatus*) .
- Inside the snails, the miracidia lose their cilia and in about 4-8 weeks successively pass through the stages of first and second generation of sporocysts. Large number of cercariae are produced by asexual reproduction with the second generation sporocyst.

- The cercariae has an elongated ovoid body about 250 μm and forked tail (furcocercus cercaria). Swarms of cercariae swim about in water for 1-3 days.
- If during that period come into contact with persons bathing or wading in water they penetrate through their unbroken skin. Skin penetration is facilitated by lytic substances secreted by penetration glands present in cercaria.

Diagram of *Schistosoma mansoni* cercaria



- ❑ On entering the skin, the cercaria shed their tails and become schistosomulae which enter the peripheral venules.
- ❑ They then start a long migration through the vena cava into the right heart , the pulmonary circulation the left heart and the systemic circulation ultimately reaching the cover liver.
- ❑ In intra hepatic portal vein, the schistosomula grow and become sexually differentiated adolescents about 20 days after skin penetration.

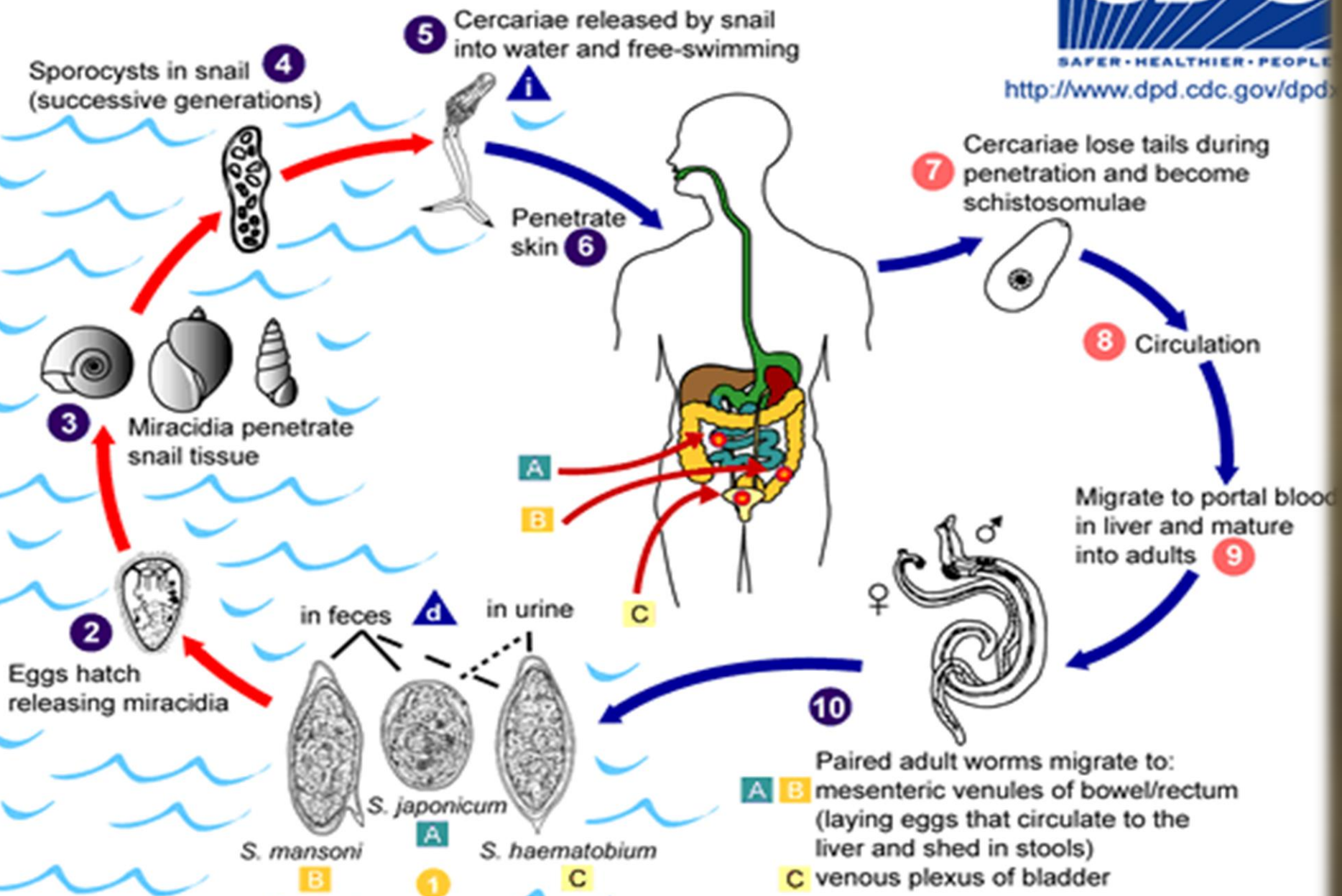
- They then start migrating against the blood stream into the inferior mesenteric veins, ultimately reaching the vesical and pelvic venus plexuses where they mature mate and begin laying eggs.
- Eggs start appearing in urine usually 10-12 weeks after cercarial penetration . the adult worms may live for 20-30 years.
- Human become the only natural definitive hosts no animal reservoir is known.

i = Infective Stage
d = Diagnostic Stage

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Pathogenicity and Clinical Features

□ Clinical illness caused by schistosomes can be classified depending on the stages in the evolution of the infection as follow:

1. Skin penetration and incubation period.
2. Egg deposition and extrusion.
3. Tissue proliferation and repair.

1.Skin penetration and incubation period.

- ❑ The clinical feature during the incubation period may be local cercarial dermatitis or general anaphylactic or toxic symptoms.
- ❑ Cercarial dermatitis consists of transient itching petechial lesions at the site of entry of the cercaria.
- ❑ This is seen more often in visitors to endemic areas than locals who may be immune due to repeated contacts. It is particularly severe when infection occurs with cercariae of non human Schistosomes (Swimmer's itch).

- ❑ Anaphylactic or toxic symptoms include fever headache, malaise and urticaria .
- ❑ This is accompanied by leucocytosis, eosinophilia, enlarged tender liver and a palpable spleen. this condition is more common in infection with *S. japonicum* (katayama fever).

2. Egg deposition and extrusion.

- ❑ The typical manifestation caused by egg laying and extrusion is painless terminal hematuria (endemic hematuria).
- ❑ Hematuria is initially microscopic, but become gross if infection heavy most patients develop frequency of micturition and burning .
- ❑ cystoscopy shows hyperplasia and inflammation of bladder mucosa with minute popular or vesicular lesions.

3. Tissue proliferation and repair.

- ❑ In chronic stage there is generalized hyperplasia and fibrosis of the vesical mucosa with granular appearance (sandy patch).
- ❑ At the site deposition of eggs , dense infiltration with lymphocytes, plasma cells, and eosinophils leads to pseudoabscesses.
- ❑ Schistosomiasis favors urinary carriage of typhoid bacilli,

- ❑ Initially trigone is involved, but ultimately the entire mucosa become inflamed, thickened and ulcerated secondary cystitis, calculi form in the bladder due to deposition of oxalate and uric crystals around the eggs and blood clots.
- ❑ There may be obstructive hyperplasia of the ureters and urethra. chronic Schistosomiasis has been associated with bladder cancer.

Diagnosis

- ❑ The eggs with characteristic terminal spine can be demonstrated by microscopic spine of centrifuged deposits of urine.
- ❑ Eggs are more abundant in the blood and pus passed by patients at the end of micturation. They also seen in the seminal fluid. They may occasionally found in feces or ,ore often in rectal or vesical biopsies.
- ❑ A refinement of diagnosis by demonstration of eggs is to hatch shed eggs into motile miracidia.

- ❑ Another diagnostic method is by detection of specific schistosome antigens in serum or urine.
- ❑ Two glycoproteins antigens associated with the gut of adult schistosomes (circulating anodic & cathodic antigens) (CAA & CCA). These can demonstrated by ELISA using monoclonal antibodies. The test is very sensitive and specific but is available only in specialized labs.
- ❑ Skin test are give positive results in all schistosomes. The intra-dermal allergic test (Fairley's test) uses antigen of cercariae, eggs and adult schistosomes from infected laboratory animals.

- ❑ Several serological tests have been described but are not very useful. These include (complement fixation), bentonite flocculation, indirect haema-agglutination, immune-fluorescence, gel diffusion & ELISA).
- ❑ Two special tests are circumoval precipitation (globular or segmented precipitation around schistosome eggs incubated in positive sera) and " cercarian-hüllen" reaction (development of pericercarial membranes around cercaria incubated in positive sera).
- ❑ Animal schistosomes can be used as antigens in these tests.

- ❑ Membrane filtration technique
- ❑ Ultrasonography is useful in diagnosing *S. haematobium*.
- ❑ Polymerase Chain Reaction (PCR).

Treatment

1. Metriphonate : drug of choice in schistosomiasis due to *haematobium*.
2. Praziquantel : is effective against all schistosomes and other trematodes and Cestodes infection.

Control

- Prophylactic measures (eradication snails)
- Prevention environmental pollution with urine and feces.
- Effective treatment of infected persons.

Schistosoma mansoni

History of distribution:

- ❑ The discovery by Manson in 1902 of eggs with lateral spine in the feces of West Indian patient, led to recognition of this second species of human schistosomes, so name *S. mansoni* .
- ❑ It widely distributed in Africa, South America and Caribbean islands.

Morphology and life cycle

- ❑ *S. mansoni* resembles *S. haematobium* in morphology and life cycle.
- ❑ The adult worm are smallest and their integument studded with prominent coarse tubercles.
- ❑ The male have a length of 6.5-10 mm it has digestive system of two caeca united in first half of the body so it has long united cecum.
- ❑ There are two suckers oral and ventral sucker and 8-9 tests close to ventral sucker are lying in a grape-like cluster.

- ❑ The female measures 7.2-14 mm, the genital organs, ovary situated in the first half of body (pre-equatorial) from which extend oviduct open in ootype. Uterus short contains a few eggs 3-5 eggs. Vitellaria or yolk glands high in number of comparison with other species extend between ootype to posterior end.
- ❑ The prepatent period (the interval between cercarial penetration and beginning of egg laying) is 4-5 weeks. The egg has a characteristic lateral spine.

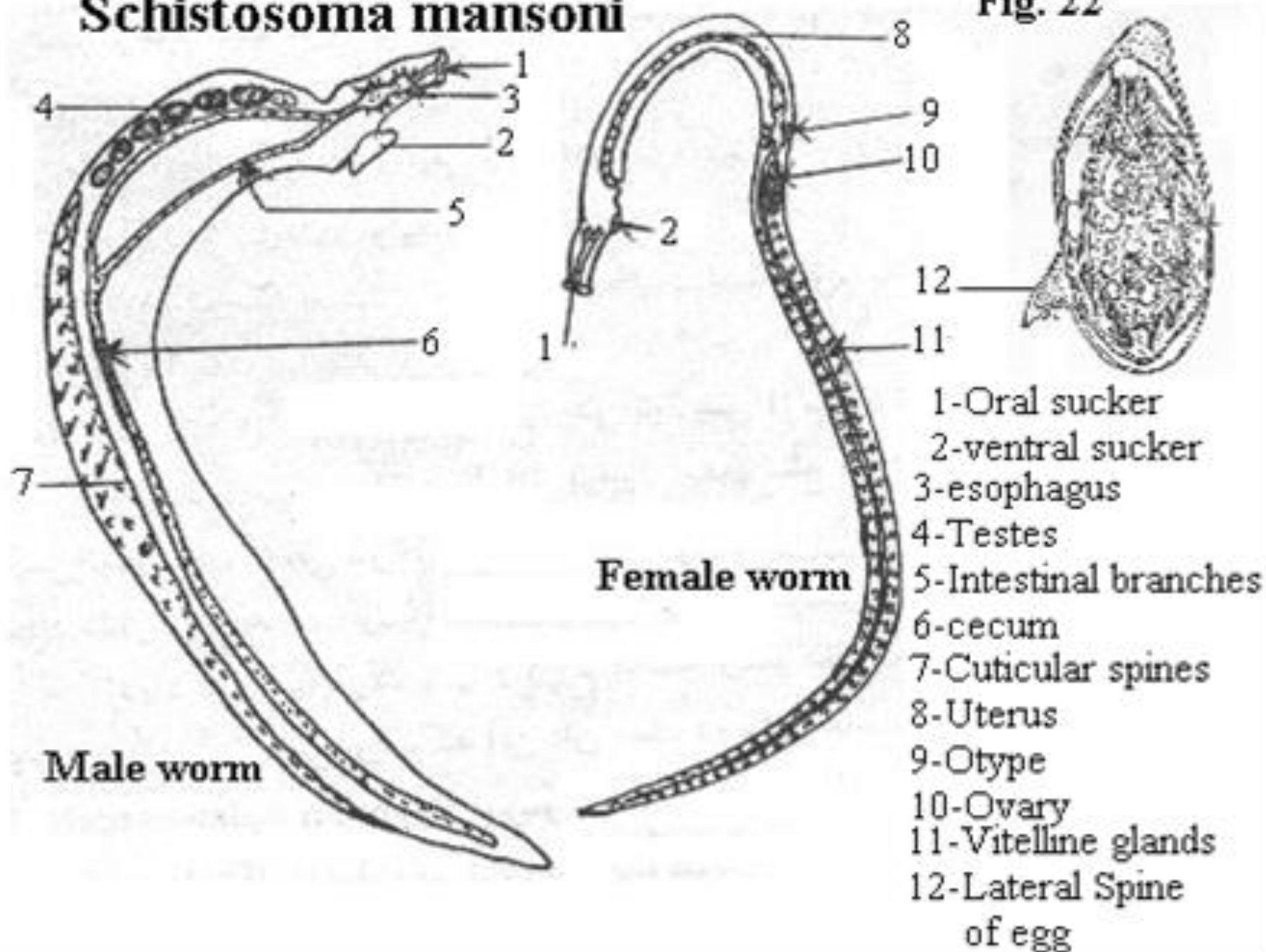
- ❑ The intermediate host are the only planorbid fresh-water snails of the genus *biomphalaria*.
- ❑ Human are only the natural definitive host through in endemic areas.
- ❑ In human the schistosomulae mature in liver and adult worms move against the blood stream into the venules of the inferior mesenteric group in the sigmoido-rectal area.
- ❑ Eggs penetrate the gut wall reach colonic lumen and are shed in feces.

Pathogenicity & clinical features

- ❑ Following skin penetration by cercariae a pruritic rash may be develop locally.
- ❑ During the stage of egg deposition the symptoms is mainly intestinal, this condition is therefore known as intestinal bilharziasis or schistosomal dysentery.

Schistosoma mansoni

Fig. 22



❑ Patients develop colicky abdominal pain and blood diarrhea which may go on intermittently for many years.

❑ The eggs deposited in the gut wall cause inflammatory reactions leading to micro-abscesses, granulomas, hyperplasia and eventual fibrosis, ectopic lesions include hepato-splenomegaly and portal hypertension. Pulmonary lesions are common in mansoni's schistosomiasis.

Diagnosis

- ☐ Eggs with lateral spine may be demonstrated microscopically in stools. Concentration methods may be required when infection is light.
- ☐ Rectal biopsies.
- ☐ Serological tests (ELISA).
- ☐ Polymerase Chain Reaction (PCR).

Treatment

- ☐ Oxamniquine is the drug of choice.
- ☐ Control:
- ☐ Control snails.
- ☐ Treatment of infected persons.
- ☐ Prevention of fecal pollution.

Schistosoma japonicum:

Distribution:

- Known as the oriental blood fluke, *S. japonicum* is found in far East, Japan, China, Taiwan and Philippines.

Morphology and life cycle:

- There are generally similar to the schistosomes described previously. The adult worms are seen in venules of the superior mesenteric veins draining the ileocaecal region.

- ❑ They are also seen in the intra-hepatic portal venules and in the haemorrhoidal plexus of veins.
- ❑ The male measures 9-22 mm and does not have cuticular tuberculation (smooth), and has two suckers oral and ventral suckers.
- ❑ The digestive system composed of two caeca united in posterior part of the body to form short united caeca. There are 6-7 testes located near ventral sucker.
- ❑ The female is long about 12-26 mm , genital organ consist of ovary which situated in middle region of parasite, the uterus filled with eggs as many as 100 eggs , vitellaria is short surrounded united caeca.

- The prepatent period 4-5 weeks. The eggs are more spherical than other species , the egg has no spine but shows lateral knob.
- Eggs deposited in the mesenteric venules penetrate gut wall and passed in feces they hatch in water and miracidia infect the intermediate host, amphibian snails of the genus *Oncomelania*.
- Man is the definitive but in endemic areas natural infection occurs in domestic animals which act as reservoirs.

Pathogenesis and clinical features:

- Cause Oriental Schistosomiasis or katayama disease, pathogenesis similar, but because probably higher egg-output; the clinical manifestation are more severe.
- The acute illness consisting of fever, abdominal pain, diarrhea and allergic manifestation is called katayama fever it is an immune complex disease caused by antibodies to the schistosomulae, adult worms and eggs.

❑ In chronic illness, liver is the site maximally affected there is initial hepatomegaly followed by fibrosis.

❑ Portal hypertension leads to oesophageal varices and gastrointestinal bleeding.

❑ The spleen is secondarily enlarged. Cerebral and pulmonary involvement may occur in some cases.

Diagnosis:

- ❑ Same demonstration of eggs in feces.

Treatment:

- ❑ *S. japonicum* more resistant to treatment than other species. A prolonged course of intravenous tartar emetric gives good results, Praziquantel also useful.

Control:

- ❑ Prevention of fecal population of soil and water.
- ❑ Treatment of infected persons.
- ❑ Snail control help to contain the infection, but presence of animal reservoirs in endemic areas makes eradication difficult.

Schistosoma intercalatum:

- ❑ First recognized in 1934 is found in West central Africa, the eggs have terminal spine, but are exclusively in stool.

Schistosoma mekongi:

- First recognized in 1978 is found in Thailand and Cambodia, along the Mekong river it is closely related to *S. japonicum*.

Thank You for attention

